



BY AMETECs

SR-04

Per-Tenant Licensing & Tenants Screen — Session Report

SmartEPT Cloud Multi-Tenancy · Session of 12 August 2026 (late evening)

Version 1.0

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1. Executive Summary

This session solved a structural licensing flaw on the shared SmartEPT cloud install (admin.smartept.com) and delivered a new operator screen, in two commits:

Repo	Commit	Content
Product (smartept)	1fcb7f7	Per-tenant licensing + Tenants (HOST) screen — 14 files, 692 insertions
Central (smartept-central)	76ddf09	Provisioning hands the tenant's licence key to the product — 1 file

The trigger: Ejaz, signed in as the product Super Admin, opened the Licence screen and saw *Khan Incorporation's* licence — another company's key, plan and seats — presented as if it were the server's own.

The real problem underneath: the product's licence store was a single row per installation. Correct for a client-hosted server (one install = one customer), wrong for the shared cloud install that hosts many tenant companies under one roof (/admin, /khan, /test1 ...). Whichever tenant's key was saved last governed *everyone*: all tenants ran on Khan's 25 device seats, the whole install's storage was reported to Central against Khan, and one tenant's expiry would have stopped every tenant's agents.

What exists now: every cloud tenant (company with `deployment_model = AMETECs_SAAS`) carries its **own** licence row — own key, own seats, own expiry, own daily phone-home, own storage reporting. Central delivers the key automatically at provisioning, so a cloud client never touches a licence screen. The Super Admin gets a new **Tenants** screen (HOST section, top of the sidebar) showing every company's licence health, seats, storage against quota, and agent activity. Client-hosted installs are deliberately untouched — their single-licence behaviour is byte-for-byte the same.

Status: committed locally on both repos; **not yet migrated, tested or deployed** — the exact steps Ejaz must run are in Section 12.

2. The Problem Ejaz Reported

Ejaz's report, in his own sequence:

1. "I am super admin of the product, right? Why was it showing Khan tenant licence on admin.smartept.com/admin?"
2. "admin@ameteecs.io is the company admin; /khan, /test1 etc. are tenants, right?"

3. "The product too should have a Tenant menu with all tenants' details — Central maintains them, but regular maintenance and their storage related to tenants would be better here too."
4. (mid-session) "Our concept itself is: Central is the super admin; the product has multiple tenants who opted for cloud subscription — but a super admin for the product is still required, right?"
5. (mid-session) "I should know the quota and how much hosting space they utilised too — it is already in Central, right?"

All five points were confirmed correct and are addressed in this build.

3. Root Cause Analysis

Where the licence lived. `installation_licenses` was created (16-Jul, R2-1) as a single-row table: *"The one licence record for this SmartEPT installation."*

`InstallationLicense::current()` returned `first()` — one slot for the whole server. Every consumer — the Licence screen, the `EnsureLicensed` agent gate, device-seat activate/release, the daily validate command, storage phone-home — read that one slot.

Why Khan's key was in it. When Khan Incorporation's cloud subscription was provisioned, there was nowhere per-tenant to put their key. The only place a key *could* be saved was the install slot — so it was, and from that moment the install identified itself to Central as Khan.

The three concrete defects this caused:

1. **Seat pooling.** "16 registered / 25 licensed" counted registered devices of *every* company on the install — Ejaz's own test devices consumed Khan's paid seats.
2. **Storage misreporting.** The phone-home reported the *whole install's* evidence storage under Khan's key, so Central's per-tenant usage stats and 90%/100% storage governance judged Khan on everyone's files.
3. **Shared fate.** Licence enforcement was install-wide: had Khan's licence expired or been suspended, agent syncing for every tenant on the server would have stopped.

The design tension. The product supports both deployment shapes: client-hosted (one company per install — singleton licence is correct) and Ametecs-hosted shared cloud (many companies — singleton is wrong). The licence model had only been built for the first shape. This session taught it the second.

4. The Concept We Locked

Agreed explicitly with Ejaz during the session:

Layer	Role	Owns
SmartEPT Central	Business super admin (Ametecs control plane)	Sales, orders, billing, invoices, plans, licence issuing, provisioning — the sole commercial truth
Product Super Admin	Host / operator of the shared install	Cross-tenant operations: storage, health, seats, activity, global SMTP, provisioning secrets, audit. Owns no licence itself
Tenant companies (khan, test1, ...)	Cloud customers	Each carries its own licence issued by Central — own key, seats, expiry

Division-of-truth rule built into the Tenants screen: the product shows what only the product knows (live storage, devices, users, sync activity, licence health); commercial records are deliberately **not** duplicated into the product — they stay in Central so there is exactly one source of truth.

5. Decisions Taken

Formal decisions (asked and answered in-session):

#	Question	Ejaz's decision
D1	Is admin.smartept.com the permanent shared home for all cloud clients?	Yes — one shared install. Per-company licensing is therefore necessary, not optional
D2	Build scope	Both together: per-company licensing + Tenants menu
D3	Khan's key currently in the install slot	Auto-move it in the migration to Khan's own tenant row
D4	Tenants menu placement	Top of the sidebar (new HOST section), Super Admin only
D5	Non-cloud companies on the shared install (/admin, tests)	Keep using the install-level slot (client-hosted parity)
D6	Perpetual clients' self-service upgrade / .lic reissue gap (raised mid-session)	Parked as a separate follow-up — see Section 14

6. What Was Built — Product (file by file)

`database/migrations/2026_08_12_000400_add_company_id_to_installation_licenses.php`
(new). Adds nullable `company_id` (unsigned bigint, **UNIQUE**) to `installation_licenses`.
 NULL = the install-level row; set = that tenant's own row (MySQL permits many NULLs under a

unique index, so both coexist). No ENUMs anywhere — house rule after four ENUM incidents.

Data move: if the install slot holds a key whose cached bundle names exactly **one**

AMETECs_SAAS company by name, the row is re-pointed to that company (Khan's case). Any ambiguity or no match → nothing moves, which makes the same migration a safe no-op on client-hosted servers.

app/Models/InstallationLicense.php . `current()` now means "the install row" (`whereNull('company_id')`), creating it on first use as before). New: `forCompany($companyId)` (`firstOrCreate` — a tenant's 7-day evaluation clock starts when its row first exists) and the single decision point **governing(?Company)** : AMETECs_SAAS company → its own row; everything else → the install row. All licence consumers now route through this one rule.

app/Services/LicenseClient.php . `validate()` was already row-based; storage phone-home now reports **only that row's company files** (install row still reports the whole box), always unscoped so the daily revalidate works under any authenticated user. The revalidate throttle lock became per-row (`license:revalidate-lock:{id}`) so one tenant's check never starves another's. `activateDevice()` / `deactivateDevice()` accept the device's `company_id` and claim/release seats against the **governing** licence's key. Central's storage-governance answer is stored per tenant (`storage_paused:{company_id}`) instead of one global switch.

app/Http/Middleware/EnsureLicensed.php . The agent gate resolves the licence governing the **requesting user's company**. A cloud tenant is judged only on its own licence; one tenant's expiry can never block another. Non-SaaS traffic behaves exactly as before.

app/Http/Controllers/Api/LicenseController.php . The Licence screen API is scope-aware: a non-super admin operates on the row governing their company; the Super Admin keeps the install row. The offline `license.lic` path (node-locked to the server's fingerprint) is install-level **only** — cloud tenant admins get a clear 403 pointing them to their key/Central. Payload gains `scope` (`company` | `installation`), `scope_company`, and per-company `devices_registered`.

app/Http/Controllers/Api/ProvisionController.php . `/api/provision` accepts a new `licence_key` : the tenant's row is created/updated with it and validated against Central immediately (best-effort — a miss is recovered by the daily validate or "Validate now"). This is the piece that makes cloud licensing zero-touch and closes the exact hole that put Khan's key in the install slot.

app/Http/Controllers/Api/DeviceController.php + **EmployeeController.php** . All three seat call-sites (register, rebind, unbind/relieve) pass the device's company so seats hit the right tenant's licence.

app/Http/Controllers/Api/ScreenshotController.php . The storage-full screenshot pause now honours both the install-wide switch and the uploading employee's per-tenant switch.

app/Console/Commands/ValidateLicense.php . The nightly (01:00) phone-home validates **every** configured row — install + each tenant — and reports each one's status; exit code fails if any is

blocked.

`app/Http/Controllers/Api/TenantController.php` (new) + `routes/api.php` . GET `/api/tenants` and POST `/api/tenants/{company}/license/validate` , both `role: SUPER_ADMIN` . The index builds the whole screen from four grouped queries (storage sum/count per company — cached 10 minutes; devices bound + last heartbeat; employees; users) plus the licence rows, and returns per company: identity (name/code/slug/status/deployment/provisioned), counts, storage used vs `storage_quota_mb` with %, licence rollup (scope own/install, status, plan, seats, expiry, last check, last error, evaluation days left), and last agent activity.

`resources/views/admin.blade.php` . New HOST nav group + Tenants item at the very top — rendered only for SUPER_ADMIN (hidden for every other role). New Tenants view: KPI chips (cloud tenants, other companies, licences OK, devices bound, total evidence storage), the tenant table with a colour storage bar (green → amber at 70% → red at 90%), per-tenant "Validate" button, branded-console /slug links, and a footer note stating the install licence status and that billing lives in Central. The Licence screen becomes scope-aware: tenant admins see "This licence belongs to — issued and managed by SmartEPT Central" and the offline-file card hides; the Super Admin sees a pointer to the Tenants screen. Full ⓘ Help entry (What / Why / How) added for the new screen.

`tests/Feature/PerTenantLicenseTest.php` (new). Five feature tests — see Section 15.

7. What Was Built — Central

`app/Services/ProductProvisioner.php` (`76ddf09`). The provision payload now includes `licence_key` = the tenant's active licence key. It is re-pushed on every provision call (slug edit, seat change, storage top-up), so the product side refreshes automatically. Nothing else in Central needed changing: `/api/v1/license/validate` already identifies the tenant by key alone, `StorageUsage` already records per-tenant daily usage, and storage governance (90% email / 100% pause) already computes per tenant — the product was simply feeding it wrong before.

8. The New Tenants Screen — Walkthrough

Location: sidebar top, **HOST** → **Tenants** (Super Admin only; other roles never see the section).

1. **KPI strip** — cloud tenant count, other companies, licences operational (x / y), total devices bound, total evidence storage on the box.
2. **Tenant table** — one row per company, cloud tenants first: name + clickable /slug (opens the branded console in a new tab), deployment model + "provisioned by Central" + retention days, ACTIVE/SUSPENDED, licence pill (ACTIVE / EVALUATION with days left / NO LICENCE / EXPIRED... with plan · kind · expiry; INSTALL pill for non-cloud companies), seats "bound / licensed" (amber when full), users, employees, storage **used of quota (%)**

with the colour bar + file count, last agent heartbeat, and a **Validate** button that forces that one tenant's phone-home right now (after a renewal or seat upgrade).

3. **Footer note** — install-level licence status, the 10-minute storage-cache notice, and the reminder that billing/orders/plans are managed in SmartEPT Central.

9. Licence Screen Behaviour by Role

Who	What the Licence screen now shows
Company Admin of a cloud tenant	Their own licence: scope line "belongs to — issued and managed by SmartEPT Central", their seats only, Save-&-validate/Validate-now act on their row. Offline-file card hidden (403 server-side)
Company Admin on a client-hosted install	Unchanged — the install licence, exactly as before
Super Admin (host)	The install-level licence (client-hosted parity) + pointer: "Cloud tenants carry their own licences — see the Tenants screen"

10. Data Migration & the Khan Key Move

On `php artisan migrate` the install slot's key moves to Khan's tenant row (name match against the cached bundle, only if exactly one AMETECs_SAAS company matches). After that:

- Khan is licensed on **their own** row — nothing to re-enter, seats now genuinely theirs.
- The **install slot becomes empty**. Non-SaaS companies on the shared install (including Ametecs' own /admin company and any LAN-model test companies) then run on a fresh **7-day evaluation** from the new empty row's creation, after which their agents block until a key is entered on the Licence screen. **Ejaz must decide** (Section 12, step 5): issue an internal licence for the install slot from Central, or ask for the operator's own company to be exempted.
- Cloud tenants without keys (test1 etc., if marked AMETECs_SAAS) each get their own 7-day evaluation, then block individually — by design, and now harmless to every other tenant.

11. Commits & Files Changed

Product 1fcb7f7 — "Per-tenant licensing on the shared cloud install + Tenants (host) screen" — 14 files, +692/-45: 3 new (TenantController, migration 2026_08_12_000400, PerTenantLicenseTest), 11 modified (InstallationLicense, LicenseClient, EnsureLicensed, LicenseController, ProvisionController, DeviceController, EmployeeController, ScreenshotController, ValidateLicense, routes/api.php, admin.blade.php).

Central 76ddf09 — *"Provisioning hands the tenant's licence key to the product"* — ProductProvisioner.php, +5.

Verification already done in-session: `php -l` clean on every PHP file; `node --check` clean on the console's 269 KB inline script; seeded company confirmed `LAN` so the existing M9 licence suite is unaffected. **Not run** (no PHP available through the bridge): the migration and the test suites — that is step 1 below.

12. What YOU Need To Do Now — Exact Instructions

Step 1 — Local verify (Laragon, `C:\laragon\www\smartept`):

```
php artisan migrate
php artisan test --filter=PerTenantLicenseTest
```

Both must be green. Then optionally the full sweep: `test.bat` (product) — M9LicenseTest and M11OffboardingTest are the ones this change could theoretically touch; they should pass unchanged.

Step 2 — Local visual check: restart Laragon (Stop All → Start All — OPcache), sign in as `super@smartept.io` → the HOST → Tenants menu appears on top; sign in as a company admin → it does not.

Step 3 — Push both repos (product: `1fcb7f7` ; central: `76ddf09`).

Step 4 — Live deploy (VPS):

```
# product (admin.smartept.com)
cd /path/to/smartept && git pull
php artisan migrate --force
php artisan config:clear && php artisan route:clear && php artisan view:clear
sudo systemctl reload php8.3-fpm

# central (smartept.com)
cd /path/to/smartept-central && git pull
php artisan view:clear
```

Step 5 — Post-deploy checks (in order):

1. Tenants screen loads; Khan row shows licence **ACTIVE** on scope "own" with their real key (auto-moved by the migration). If it shows EVALUATION instead, press **Validate** on Khan's row; if still unconfigured, re-run provisioning for Khan from Central (Save on the tenant) — Central now pushes the key.

2. Super Admin → Licence screen now shows the **install** slot (likely EVALUATION after the move) — **decide**: issue an internal Ametecs licence for the install slot, or tell me to exempt the operator's own company. Until then agents of non-SaaS companies stop after 7 days.
3. A Khan agent signs in and syncs (their own licence now gates them).
4. Central → tenant storage stats: from tomorrow's phone-homes each tenant's `storage_gb` is its own, not the whole box.
5. Provision a fresh test tenant from Central end-to-end: company + admin + **licence auto-active** with zero manual key entry.

Step 6 — run BF-01 journeys if you want the full business-flow sign-off, and execute SR-05 (the test suite shipped with this report).

13. Risks & Important Notes

1. **The install-slot evaluation window** (Section 10) is the one behavioural side-effect on live. It is deliberate (D5) but needs your Step-5.2 decision within 7 days of migrating.
2. The migration's key move is name-match based (bundle company name = company name, exactly one match). If Khan's product-side company was renamed, the move no-ops — then one "Validate" click after Central re-provisions fixes it. Nothing can be moved to a *wrong* tenant: ambiguity aborts the move.
3. Storage figures on the Tenants screen are cached 10 minutes (`tenants:storage-by-company`) — a heavy SUM over evidence files; the cache note is printed on-screen.
4. `storage_paused` handling is now two-level (global + per-tenant). Existing global pauses keep working; per-tenant pauses arrive with each tenant's own phone-home.
5. Central TEST-MODE signup OTP oddity from earlier today remains uninvestigated (unrelated, noted for completeness).
6. Agent copies' divergence re `76b96be` (meetings) is untouched by this session.

14. Deferred — Agreed Follow-Up Backlog

Perpetual / client-hosted self-service licensing (Ejaz, this session): a perpetual client today has **no** self-service way to upgrade/downgrade a plan or add seats — and Central has no workflow to reissue an updated node-locked `license.lic`. It is a manual WhatsApp/email affair. Agreed scope for a future session: client-portal "request upgrade/seats" flow → Central admin approve + reissue `.lic` (existing RSA-2048 signer) → product notice to import the new file. Not started, by agreement — one thing at a time.

15. Test Coverage

Automated (shipped, in `tests/Feature/PerTenantLicenseTest.php`): governing-row resolution (SaaS → own, others/null → install); SaaS company admin saves & reads *their* key with the install slot untouched, scope-marked payload; SaaS admin blocked (403) from offline `.lic` import; `/api/provision` with `licence_key` creating an **active** tenant row hands-free; `/api/tenants` denied to company admin, correct per-tenant rollup for super.

Manual: SR-05 — *Per-Tenant Licensing & Tenants Screen Test Case Suite* (Excel workbook + PDF), 58 cases across 10 areas with per-case steps, expected results, priority and a result-tracking column. Execute after Step 4; log any P1 failure to me immediately.

Prepared by Claude (Cowork) for Ametecs Pvt Ltd — internal engineering report. Commercial truth for licences remains SmartEPT Central.